

Proyek A/B Testing

Proyek ini berisi analisis hasil pengujian A/B dari sebuah toko daring internasional.

Pendahuluan

Anda telah menerima sebuah tugas analitik dari suatu toko daring internasional. Pendahulu Anda gagal untuk menyelesaikan tugas ini: dia sempat menjalankan A/B testing sebelum akhirnya berhenti dari pekerjaannya (katanya sih, demi beternak ayam di kampung). Dia pun hanya meninggalkan spesifikasi teknis dan hasil dari eksperimen yang dilakukannya.

Tujuan

- Memeriksa perubahan yang terkait dengan pengenalan sistem rekomendasi yang ditingkatkan.
- Mengetahui validitas hasil tes dan metodenya.
- Menganalisis dan mendapatkan wawasan dari hasil tes.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as st
import plotly.graph_objects as go
import math

import warnings
warnings.filterwarnings('ignore')
```

Observasi Data

Load Dataset

```
In [2]: event_time = pd.read_csv('/datasets/ab_project_marketing_events_us.csv')
users = pd.read_csv('/datasets/final_ab_new_users_upd_us.csv')
events = pd.read_csv('/datasets/final_ab_events_upd_us.csv')
participant = pd.read_csv('/datasets/final_ab_participants_upd_us.csv')
```

Tabel 'event_time'

```
In [3]: event_time.head()
```

Out[3]:

	name	regions	start_dt	finish_dt
0	Christmas&New Year Promo	EU, N.America	2020-12-25	2021-01-03
1	St. Valentine's Day Giveaway	EU, CIS, APAC, N.America	2020-02-14	2020-02-16
2	St. Patrick's Day Promo	EU, N.America	2020-03-17	2020-03-19
3	Easter Promo	EU, CIS, APAC, N.America	2020-04-12	2020-04-19
4	4th of July Promo	N.America	2020-07-04	2020-07-11

In [4]: `event_time.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14 entries, 0 to 13
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   name         14 non-null    object
1   regions      14 non-null    object
2   start_dt     14 non-null    object
3   finish_dt    14 non-null    object
dtypes: object(4)
memory usage: 576.0+ bytes
```

In [5]: `event_time['start_dt'] = event_time['start_dt'].astype('datetime64[ns]')`
`event_time['finish_dt'] = event_time['finish_dt'].astype('datetime64[ns]')`
`event_time.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14 entries, 0 to 13
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   name         14 non-null    object
1   regions      14 non-null    object
2   start_dt     14 non-null    datetime64[ns]
3   finish_dt    14 non-null    datetime64[ns]
dtypes: datetime64[ns](2), object(2)
memory usage: 576.0+ bytes
```

In [6]: `event_time.isna().sum()`

Out[6]:

```
name         0
regions      0
start_dt     0
finish_dt    0
dtype: int64
```

Pada tabel ini terdapat tipe data yang tidak sesuai pada kolo, start_ts dan finish_ts yang mana seharusnya tipe data pada kolom tersebut adalah datetime. Maka dari itu perlu dilakukan perubahan tipe data pada kolom tersebut.

Tabel 'users'

In [7]: `users.head(10)`

Out[7]:

	user_id	first_date	region	device
0	D72A72121175D8BE	2020-12-07	EU	PC
1	F1C668619DFE6E65	2020-12-07	N.America	Android
2	2E1BF1D4C37EA01F	2020-12-07	EU	PC
3	50734A22C0C63768	2020-12-07	EU	iPhone
4	E1BDDCE0DAFA2679	2020-12-07	N.America	iPhone
5	137119F5A9E69421	2020-12-07	N.America	iPhone
6	62F0C741CC42D0CC	2020-12-07	APAC	iPhone
7	8942E64218C9A1ED	2020-12-07	EU	PC
8	499AFACF904BBAE3	2020-12-07	N.America	iPhone
9	FFCEA1179C253104	2020-12-07	EU	Android

In [8]: `users.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 58703 entries, 0 to 58702
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   user_id     58703 non-null  object
1   first_date  58703 non-null  object
2   region      58703 non-null  object
3   device      58703 non-null  object
dtypes: object(4)
memory usage: 1.8+ MB
```

In [9]: `users['first_date'] = users['first_date'].astype('datetime64[ns]')`
`users.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 58703 entries, 0 to 58702
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   user_id     58703 non-null  object
1   first_date  58703 non-null  datetime64[ns]
2   region      58703 non-null  object
3   device      58703 non-null  object
dtypes: datetime64[ns](1), object(3)
memory usage: 1.8+ MB
```

In [10]: `users.duplicated().sum()`

Out[10]: 0

In [11]: `users['region'].unique()`

Out[11]: array(['EU', 'N.America', 'APAC', 'CIS'], dtype=object)

In [12]: `users['device'].unique()`

Out[12]: array(['PC', 'Android', 'iPhone', 'Mac'], dtype=object)

In [13]: `users.isna().sum()`

```
Out[13]: user_id      0
         first_date   0
         region       0
         device       0
         dtype: int64
```

Kolom first_date pda tabel users seharusnya memiliki tipe data datetime, untuk itu akan diubah tipe datanya.

Tabel 'events'

```
In [14]: events.head(10)
```

```
Out[14]:
```

	user_id	event_dt	event_name	details
0	E1BDDCE0DAFA2679	2020-12-07 20:22:03	purchase	99.99
1	7B6452F081F49504	2020-12-07 09:22:53	purchase	9.99
2	9CD9F34546DF254C	2020-12-07 12:59:29	purchase	4.99
3	96F27A054B191457	2020-12-07 04:02:40	purchase	4.99
4	1FD7660FDF94CA1F	2020-12-07 10:15:09	purchase	4.99
5	831887FE7F2D6CBA	2020-12-07 06:50:29	purchase	4.99
6	6B2F726BFD5F8220	2020-12-07 11:27:42	purchase	4.99
7	BEB37715AACF53B0	2020-12-07 04:26:15	purchase	4.99
8	B5FA27F582227197	2020-12-07 01:46:37	purchase	4.99
9	A92195E3CFB83DBD	2020-12-07 00:32:07	purchase	4.99

```
In [15]: events.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 423761 entries, 0 to 423760
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   user_id     423761 non-null  object
1   event_dt    423761 non-null  object
2   event_name  423761 non-null  object
3   details     60314 non-null   float64
dtypes: float64(1), object(3)
memory usage: 12.9+ MB
```

```
In [16]: events['event_name'].unique()
```

```
Out[16]: array(['purchase', 'product_cart', 'product_page', 'login'], dtype=object)
```

```
In [17]: events.loc[events['details'].isna()]
```

Out[17]:

		user_id	event_dt	event_name	details
	60314	2E1BF1D4C37EA01F	2020-12-07 09:05:47	product_cart	NaN
	60315	50734A22C0C63768	2020-12-07 13:24:03	product_cart	NaN
	60316	5EB159DA9DC94DBA	2020-12-07 22:54:02	product_cart	NaN
	60317	084A22B980BA8169	2020-12-07 15:25:55	product_cart	NaN
	60318	0FC21E6F8FAA8DEC	2020-12-07 06:56:27	product_cart	NaN
	...	...	...	...	...
	423756	245E85F65C358E08	2020-12-30 19:35:55	login	NaN
	423757	9385A108F5A0A7A7	2020-12-30 10:54:15	login	NaN
	423758	DB650B7559AC6EAC	2020-12-30 10:59:09	login	NaN
	423759	F80C9BDDEA02E53C	2020-12-30 09:53:39	login	NaN
	423760	7AEC61159B672CC5	2020-12-30 11:36:13	login	NaN

363447 rows × 4 columns

In [18]: `events.loc[(events['event_name']=='purchase')].shape[0]`

Out[18]: 60314

In [19]: `events['event_dt'] = events['event_dt'].astype('datetime64[ns]')`
`events.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 423761 entries, 0 to 423760
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   user_id     423761 non-null object
1   event_dt    423761 non-null datetime64[ns]
2   event_name  423761 non-null object
3   details     60314 non-null float64
dtypes: datetime64[ns](1), float64(1), object(2)
memory usage: 12.9+ MB
```

In [20]: `events.duplicated().sum()`

Out[20]: 0

In [21]: `events['event_dt'].min()`

Out[21]: Timestamp('2020-12-07 00:00:33')

In [22]: `events['event_dt'].max()`

Out[22]: Timestamp('2020-12-30 23:36:33')

In [23]: `events.isna().sum()`

```
Out[23]: user_id      0
         event_dt    0
         event_name   0
         details     363447
         dtype: int64
```

```
In [24]: (events.isna().sum()/len(events))*100
```

```
Out[24]: user_id      0.000000
         event_dt    0.000000
         event_name   0.000000
         details     85.766977
         dtype: float64
```

Dari hasil pengamatan pada tabel events, didapatkan tipe data yang salah pada kolom event_dt, sehingga harus diubah ke tipe data yang sesuai. Hasil pengamatan lainnya dapat disimpulkan bahwa tidak ada hal yang aneh lagi pada tabel event. Selain itu juga terdapat nilai null pada tabel events, yang mana nilai null nya sebesar 85% dari keseluruhan data. Hal ini mengakibatkan kita tidak bisa membuang nilai null tersebut, nilai null selanjutnya akan diisi dengan metode yang sesuai.

Tabel 'participant'

```
In [25]: participant.head(10)
```

```
Out[25]:
```

	user_id	group	ab_test
0	D1ABA3E2887B6A73	A	recommender_system_test
1	A7A3664BD6242119	A	recommender_system_test
2	DABC14FDDFADD29E	A	recommender_system_test
3	04988C5DF189632E	A	recommender_system_test
4	4FF2998A348C484F	A	recommender_system_test
5	7473E0943673C09E	A	recommender_system_test
6	C46FE336D240A054	A	recommender_system_test
7	92CB588012C10D3D	A	recommender_system_test
8	B3A2485649E4A012	A	recommender_system_test
9	66FC298441D50783	A	recommender_system_test

```
In [26]: participant.isna().sum()
```

```
Out[26]: user_id      0
         group       0
         ab_test     0
         dtype: int64
```

```
In [27]: participant['group'].unique()
```

```
Out[27]: array(['A', 'B'], dtype=object)
```

```
In [28]: participant['ab_test'].unique()
```

```
Out[28]: array(['recommender_system_test', 'interface_eu_test'], dtype=object)
```

```
In [29]: participant.duplicated().sum()
```

```
Out[29]: 0
```

Tabel participant tidak memiliki kesalahan dalam tipe data dan tidak terdapat nilai duplikat maupun yang hilang di dalamnya.

Pra-Pemrosesan Data

Mengisi Nilai Yang Hilang

Dalam tabel events, terdapat nilai yang hilang sebanyak 85% pada kolom details. Karena banyaknya data yang hilang, maka data yang hilang akan diisi dengan metode yang sesuai.

```
In [30]: events.loc[(events['details'].isna()).head(5)]
```

```
Out[30]:
```

	user_id	event_dt	event_name	details
60314	2E1BF1D4C37EA01F	2020-12-07 09:05:47	product_cart	NaN
60315	50734A22C0C63768	2020-12-07 13:24:03	product_cart	NaN
60316	5EB159DA9DC94DBA	2020-12-07 22:54:02	product_cart	NaN
60317	084A22B980BA8169	2020-12-07 15:25:55	product_cart	NaN
60318	0FC21E6F8FAA8DEC	2020-12-07 06:56:27	product_cart	NaN

```
In [31]: events['event_name'].unique()
```

```
Out[31]: array(['purchase', 'product_cart', 'product_page', 'login'], dtype=object)
```

```
In [32]: events.loc[events['details'].isna(), 'event_name'].unique()
```

```
Out[32]: array(['product_cart', 'product_page', 'login'], dtype=object)
```

```
In [33]: events.loc[~(events['details'].isna()), 'event_name'].unique()
```

```
Out[33]: array(['purchase'], dtype=object)
```

Jika melihat nilai unik dari kolom event_name, didapatkan kesimpulan bahwa ketika kolom data tidak memiliki nilai, kolom tersebut hanya ada saat pengguna melakukan pembelian. Hal ini menandakan bahwa memang hanya proses pembelian saja yang memiliki nilai pada kolom tersebut. Maka dari itu nilai yang hilang bisa diganti dengan angka '0'.

```
In [34]: events.loc[events['details'].isna(), 'details'] = 0
```

```
In [35]: events.isna().sum()
```

```
Out[35]: user_id      0
event_dt      0
event_name     0
details        0
dtype: int64
```

Filter Data

Untuk mengetahui pengguna yang mengalami peningkatan dari konversi ke halaman selanjutnya dalam kurun waktu 14 hari, maka akan difilter umur penggunanya. Untuk mempermudah perhitungan, dibuat kolom baru yang berisi tanggal dimana pengguna mulai memasuki halaman.

```
In [36]: events['date'] = events['event_dt'].dt.date.astype('datetime64')
```

```
In [37]: users_eu = users.loc[users['region']=='EU']
```

```
In [38]: users_eu.shape
```

```
Out[38]: (43396, 4)
```

```
In [39]: events_user_eu = pd.merge(events, users_eu, on=['user_id'], how='outer')
events_user_eu
```

Out[39]:

		user_id	event_dt	event_name	details	date	first_date	region	device
0	E1BDDCE0DAFA2679		2020-12-07 20:22:03	purchase	99.99	2020-12-07	NaT	NaN	NaN
1	E1BDDCE0DAFA2679		2020-12-09 06:21:35	purchase	9.99	2020-12-09	NaT	NaN	NaN
2	E1BDDCE0DAFA2679		2020-12-07 20:22:03	login	0.00	2020-12-07	NaT	NaN	NaN
3	E1BDDCE0DAFA2679		2020-12-09 06:21:35	login	0.00	2020-12-09	NaT	NaN	NaN
4	7B6452F081F49504		2020-12-07 09:22:53	purchase	9.99	2020-12-07	2020-12-07	EU	iPhone
...	...	...	...	...	...	...	...	...	...
423756	C9F1399FD83F13		2020-12-24 04:47:08	login	0.00	2020-12-24	2020-12-23	EU	iPhone
423757	C9F1399FD83F13		2020-12-27 09:47:46	login	0.00	2020-12-27	2020-12-23	EU	iPhone
423758	4F32B58E4FF19DC5		2020-12-23 20:01:29	login	0.00	2020-12-23	2020-12-23	EU	Android
423759	27767D4BECDD260E		2020-12-26 19:02:18	login	0.00	2020-12-26	2020-12-07	EU	iPhone
423760	27767D4BECDD260E		2020-12-27 04:57:22	login	0.00	2020-12-27	2020-12-07	EU	iPhone

423761 rows × 8 columns

```
In [40]: events_user_eu['since_start'] = events_user_eu['date'] - events_user_eu['first_date']

In [41]: events_user_eu = events_user_eu.loc[events_user_eu['since_start']<='14 days']
events_user_eu
```

Out[41]:

		user_id	event_dt	event_name	details	date	first_date	region	device	si
4	7B6452F081F49504		2020-12-07 09:22:53	purchase	9.99	2020-12-07	2020-12-07	EU	iPhone	
5	7B6452F081F49504		2020-12-09 01:50:37	purchase	4.99	2020-12-09	2020-12-07	EU	iPhone	
6	7B6452F081F49504		2020-12-21 08:04:05	purchase	4.99	2020-12-21	2020-12-07	EU	iPhone	
8	7B6452F081F49504		2020-12-07 09:22:55	product_page	0.00	2020-12-07	2020-12-07	EU	iPhone	
9	7B6452F081F49504		2020-12-09 01:50:38	product_page	0.00	2020-12-09	2020-12-07	EU	iPhone	
...	...	...	...	...	...	...	...	...	...	...
423752	85500F5DD194ADBA		2020-12-23 09:33:24	login	0.00	2020-12-23	2020-12-23	EU	Android	
423755	C9F1399FD83F13		2020-12-23 15:04:28	login	0.00	2020-12-23	2020-12-23	EU	iPhone	
423756	C9F1399FD83F13		2020-12-24 04:47:08	login	0.00	2020-12-24	2020-12-23	EU	iPhone	
423757	C9F1399FD83F13		2020-12-27 09:47:46	login	0.00	2020-12-27	2020-12-23	EU	iPhone	
423758	4F32B58E4FF19DC5		2020-12-23 20:01:29	login	0.00	2020-12-23	2020-12-23	EU	Android	

304714 rows × 9 columns



Untuk saat ini telah didapatkan pengguna yang melakukan kegiatan dalam kurun waktu 14 hari.

EDA

Distribusi Data

In [42]:

```
events_user_eu['event_dt'].max()
```

Out[42]:

```
Timestamp('2020-12-29 23:59:36')
```

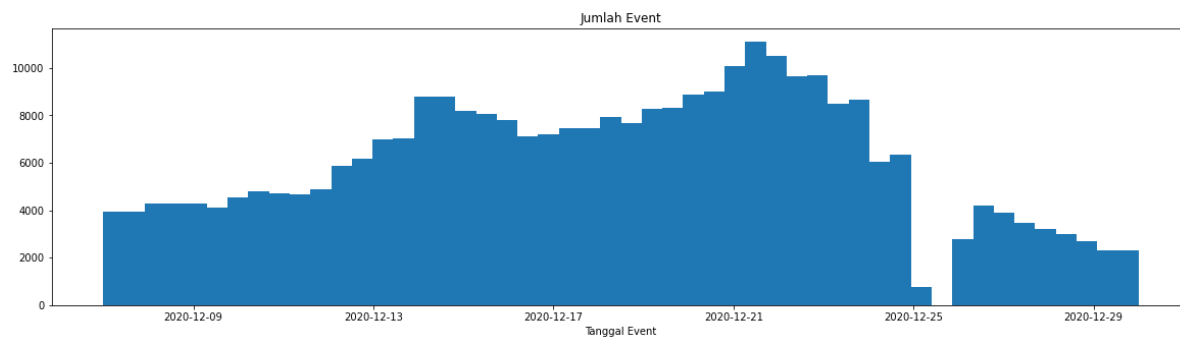
In [43]:

```
events_user_eu['event_dt'].min()
```

Out[43]:

```
Timestamp('2020-12-07 00:00:33')
```

```
In [44]: plt.figure(figsize=(20,5))  
  
plt.hist(events_user_eu['event_dt'], bins=50)  
  
plt.title('Jumlah Event')  
  
plt.xlabel('Tanggal Event')  
  
plt.show()
```



```
In [45]: events_user_eu.groupby('date').agg({'user_id': 'count'})
```

Out[45]:

	user_id
date	
2020-12-07	8520
2020-12-08	9327
2020-12-09	9136
2020-12-10	10422
2020-12-11	10258
2020-12-12	13053
2020-12-13	15374
2020-12-14	19415
2020-12-15	17422
2020-12-16	15609
2020-12-17	16091
2020-12-18	16980
2020-12-19	18055
2020-12-20	19608
2020-12-21	24064
2020-12-22	21234
2020-12-23	18798
2020-12-24	13455
2020-12-26	8998
2020-12-27	7614
2020-12-28	6240
2020-12-29	5041

Dari hasil distribusi data dan grafik, dapat dilihat tidak adanya kegiatan yang terjadi di tanggal 25.

```
In [46]: (len(events_user_eu.loc[events_user_eu['date']>'2020-12-25'])/len(events_user_eu))
```

```
Out[46]: 9.153829492573363
```

Jumlah pengguna setelah tanggal 25 adalah sebanyak 9% dari data awal, hal ini memungkinkan kita untuk mengambil data sampai tanggal 24 saja atau bisa juga mengambil keseluruhan data.

```
In [47]: events_fix = events_user_eu.loc[~(events_user_eu['date']>'2020-12-25')]
events_fix
```

Out[47]:

		user_id	event_dt	event_name	details	date	first_date	region	device	s
4	7B6452F081F49504		2020-12-07 09:22:53	purchase	9.99	2020-12-07	2020-12-07	EU	iPhone	
5	7B6452F081F49504		2020-12-09 01:50:37	purchase	4.99	2020-12-09	2020-12-07	EU	iPhone	
6	7B6452F081F49504		2020-12-21 08:04:05	purchase	4.99	2020-12-21	2020-12-07	EU	iPhone	
8	7B6452F081F49504		2020-12-07 09:22:55	product_page	0.00	2020-12-07	2020-12-07	EU	iPhone	
9	7B6452F081F49504		2020-12-09 01:50:38	product_page	0.00	2020-12-09	2020-12-07	EU	iPhone	
...	...	...	...	...	...	...	...	...	...	...
423747	8523DDD73A2CB208		2020-12-24 05:40:16	login	0.00	2020-12-24	2020-12-23	EU	PC	
423752	85500F5DD194ADBA		2020-12-23 09:33:24	login	0.00	2020-12-23	2020-12-23	EU	Android	
423755	C9F1399FDCD83F13		2020-12-23 15:04:28	login	0.00	2020-12-23	2020-12-23	EU	iPhone	
423756	C9F1399FDCD83F13		2020-12-24 04:47:08	login	0.00	2020-12-24	2020-12-23	EU	iPhone	
423758	4F32B58E4FF19DC5		2020-12-23 20:01:29	login	0.00	2020-12-23	2020-12-23	EU	Android	

276821 rows × 9 columns



Funnel

Secara penjelasan, untuk urutan tahap pada funnel adalah sebagai berikut:

1. Login
2. Product Page
3. Product Cart
4. Purchase

In [48]:

```
events_fix.head()
```

Out[48]:

		user_id	event_dt	event_name	details	date	first_date	region	device	since_start
4	7B6452F081F49504		2020-12-07 09:22:53	purchase	9.99	2020-12-07	2020-12-07	EU	iPhone	0 days
5	7B6452F081F49504		2020-12-09 01:50:37	purchase	4.99	2020-12-09	2020-12-07	EU	iPhone	2 days
6	7B6452F081F49504		2020-12-21 08:04:05	purchase	4.99	2020-12-21	2020-12-07	EU	iPhone	14 days
8	7B6452F081F49504		2020-12-07 09:22:55	product_page	0.00	2020-12-07	2020-12-07	EU	iPhone	0 days
9	7B6452F081F49504		2020-12-09 01:50:38	product_page	0.00	2020-12-09	2020-12-07	EU	iPhone	2 days

```
In [49]: user = events_fix.pivot_table(
        index='user_id',
        columns='event_name',
        values='event_dt',
        aggfunc='min')
```

```
In [50]: step_1 = ~user['login'].isna()
step_2 = step_1 & (user['product_page'] > user['login'])
step_3 = step_2 & (user['product_cart'] > user['product_page'])
step_4 = step_3 & (user['purchase'] > user['product_cart'])

login = user[step_1].shape[0]
product_page = user[step_2].shape[0]
product_cart = user[step_3].shape[0]
purchase = user[step_4].shape[0]
```

```
In [51]: arr = np.array([[login], [product_page], [product_cart], [purchase]])
name = np.array(['Login', 'Product Page', 'Product Cart', 'Purchase'])

funnel = pd.DataFrame({
    'event_name': name,
    'quantity': arr.flatten()
}, columns = ['event_name', 'quantity'])
```

```
In [52]: for i in range(len(funnel)):
        if i == 0:
            funnel['total_conversion'] = (funnel['quantity'] / funnel['quantity']).loc[i]
            funnel['conversion_to_nextstep'] = (funnel['total_conversion'] / funnel['total_conversion']).loc[i]
        else:
            funnel.loc[i, 'total_conversion'] = (funnel['quantity'].loc[i] / funnel['quantity'].loc[0])
            funnel.loc[i, 'conversion_to_nextstep'] = (funnel['total_conversion'].loc[i] / funnel['total_conversion'].loc[0])
```

```
In [53]: funnel['total_conversion'] = round(funnel['total_conversion']).astype('int')
funnel['conversion_to_nextstep'] = round(funnel['conversion_to_nextstep']).astype('int')
```

```
In [54]: funnel
```

Out[54]:

	event_name	quantity	total_conversion	conversion_to_nextstep
0	Login	43381	100	100
1	Product Page	16311	38	38
2	Product Cart	981	6	16
3	Purchase	5	1	8

In [55]:

```
fig = go.Figure(go.Funnel(  
    name = 'Conversion Funnel',  
    y = funnel['event_name'],  
    x = funnel['quantity'],  
    textposition = 'inside',  
    textinfo = 'value+percent previous',  
    marker = {"color": ["deepskyblue", "lightsalmon", "tan", "teal", "wheat"]}  
))  
  
fig.update_layout(  
    title = {'text': 'Konversi Corong Peristiwa', 'x': 0.5},  
    margin = {'l': 40, 'r': 200},  
    font = {'size': 12}  
)  
  
fig.show()
```

Konversi (



Dari hasil funnel, kita dapat menyimpulkan bahwa tahap login merupakan tahap dimana aktifitas terbanyak ditemukan, yaitu sebanyak 43381 aktifitas. Dari keseluruhan jumlah

pengguna, hanya 38% yang melanjutkan ke tahap product page.

Pada tahap product page, memiliki jumlah pengunjung sebanyak 16311 pelanggan, yang merupakan sekitar 26% dari total pengunjung. Dari pengunjung tahap ini, hanya 16% yang melanjutkan ke tahap product cart.

Tahap product cart memiliki jumlah pengguna yang paling sedikit yaitu 981.

Dari funnel dapat disimpulkan hal ini tidak sesuai dengan apa yang diharapkan dimana disetiap tahapan minimal akan ada peningkatan sebesar 10%.

A/B Testing

Pengujian A/B Testing yang akan diambil adalah recommender_system_test. Untuk itu data akan difilter terlebih dahulu.

```
In [56]: participant['ab_test'].value_counts()
```

```
Out[56]: interface_eu_test      10850  
recommender_system_test      3675  
Name: ab_test, dtype: int64
```

```
In [57]: participant_new = participant.loc[participant['ab_test']=='recommender_system_test']  
participant_new
```

```
Out[57]:
```

	user_id	group	ab_test
0	D1ABA3E2887B6A73	A	recommender_system_test
1	A7A3664BD6242119	A	recommender_system_test
2	DABC14FDDFADD29E	A	recommender_system_test
3	04988C5DF189632E	A	recommender_system_test
4	4FF2998A348C484F	A	recommender_system_test
...	...	...	...
3670	053FB26D6D49EDDC	A	recommender_system_test
3671	9D263B8EF15CF188	B	recommender_system_test
3672	F2FBBA33F37DEC46	A	recommender_system_test
3673	29C92313A98B1176	B	recommender_system_test
3674	6715343AFBA285AE	B	recommender_system_test

3675 rows × 3 columns

```
In [58]: events_fix.head()
```

Out[58]:

		user_id	event_dt	event_name	details	date	first_date	region	device	since_start
4	7B6452F081F49504		2020-12-07 09:22:53	purchase	9.99	2020-12-07	2020-12-07	EU	iPhone	0 days
5	7B6452F081F49504		2020-12-09 01:50:37	purchase	4.99	2020-12-09	2020-12-07	EU	iPhone	2 days
6	7B6452F081F49504		2020-12-21 08:04:05	purchase	4.99	2020-12-21	2020-12-07	EU	iPhone	14 days
8	7B6452F081F49504		2020-12-07 09:22:55	product_page	0.00	2020-12-07	2020-12-07	EU	iPhone	0 days
9	7B6452F081F49504		2020-12-09 01:50:38	product_page	0.00	2020-12-09	2020-12-07	EU	iPhone	2 days

In [59]:

events_fix.shape

Out[59]:

(276821, 9)

In [60]:

participant_new.head()

Out[60]:

	user_id	group	ab_test
0	D1ABA3E2887B6A73	A	recommender_system_test
1	A7A3664BD6242119	A	recommender_system_test
2	DABC14FDDFADD29E	A	recommender_system_test
3	04988C5DF189632E	A	recommender_system_test
4	4FF2998A348C484F	A	recommender_system_test

In [61]:

participant_new.shape

Out[61]:

(3675, 3)

In [62]:

df = pd.merge(events_fix, participant_new, on=['user_id'], how='inner')
df

Out[62]:

		user_id	event_dt	event_name	details	date	first_date	region	device	sinc
0	831887FE7F2D6CBA		2020-12-07 06:50:29	purchase	4.99	2020-12-07	2020-12-07	EU	Android	
1	831887FE7F2D6CBA		2020-12-09 02:19:17	purchase	99.99	2020-12-09	2020-12-07	EU	Android	
2	831887FE7F2D6CBA		2020-12-07 06:50:30	product_cart	0.00	2020-12-07	2020-12-07	EU	Android	
3	831887FE7F2D6CBA		2020-12-08 10:52:27	product_cart	0.00	2020-12-08	2020-12-07	EU	Android	
4	831887FE7F2D6CBA		2020-12-09 02:19:17	product_cart	0.00	2020-12-09	2020-12-07	EU	Android	
...	...	...	...	...	...	...	...	...	...	...
20112	1484BBF124DB1B18		2020-12-22 16:02:18	login	0.00	2020-12-22	2020-12-21	EU	PC	
20113	BEF16764A13AEC34		2020-12-21 03:49:49	login	0.00	2020-12-21	2020-12-21	EU	PC	
20114	BEF16764A13AEC34		2020-12-22 18:52:25	login	0.00	2020-12-22	2020-12-21	EU	PC	
20115	BEF16764A13AEC34		2020-12-24 22:11:00	login	0.00	2020-12-24	2020-12-21	EU	PC	
20116	23DDD27AC3FEFA63		2020-12-21 02:51:45	login	0.00	2020-12-21	2020-12-21	EU	PC	

20117 rows × 11 columns

In [63]:

```
df_group = df.groupby('group').agg({'user_id': 'nunique'}).reset_index()
df_group
```

Out[63]:

	group	user_id
0	A	2604
1	B	877

Didapatkan jumlah untuk A/B testing, group A memiliki jumlah terbesar yaitu 2604 pengguna sedangkan group B sebanyak 877 pengguna.

Kemungkinan Pengguna Berada Di Kedua Grup

In [64]:

```
check_users = df.groupby('user_id')['group'].nunique()
```

```
In [65]: check_users[check_users > 1].count()
```

```
Out[65]: 0
```

Tidak ada pengguna yang berada di kedua sampel.

Distribusi Jumlah Data Per Hari

```
In [66]: df.head()
```

```
Out[66]:
```

	user_id	event_dt	event_name	details	date	first_date	region	device	since_stai
0	831887FE7F2D6CBA	2020-12-07 06:50:29	purchase	4.99	2020-12-07	2020-12-07	EU	Android	0 day
1	831887FE7F2D6CBA	2020-12-09 02:19:17	purchase	99.99	2020-12-09	2020-12-07	EU	Android	2 day
2	831887FE7F2D6CBA	2020-12-07 06:50:30	product_cart	0.00	2020-12-07	2020-12-07	EU	Android	0 day
3	831887FE7F2D6CBA	2020-12-08 10:52:27	product_cart	0.00	2020-12-08	2020-12-07	EU	Android	1 day
4	831887FE7F2D6CBA	2020-12-09 02:19:17	product_cart	0.00	2020-12-09	2020-12-07	EU	Android	2 day

```
In [67]: event_day = df.groupby(['date']).agg({'event_dt': 'count'})

event_day = event_day.rename(columns={'event_dt': 'event_count'})

event_day
```

Out[67]:

	event_count
date	
2020-12-07	674
2020-12-08	551
2020-12-09	709
2020-12-10	580
2020-12-11	517
2020-12-12	545
2020-12-13	472
2020-12-14	1263
2020-12-15	1252
2020-12-16	1376
2020-12-17	1481
2020-12-18	1503
2020-12-19	1718
2020-12-20	1743
2020-12-21	2304
2020-12-22	1386
2020-12-23	1116
2020-12-24	927

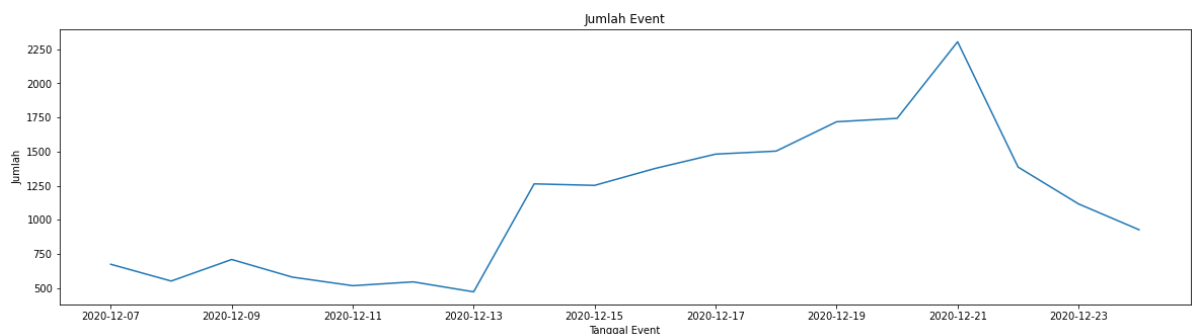
```
In [68]: plt.figure(figsize=(20,5))

plt.plot(event_day.index, event_day['event_count'])

plt.title('Jumlah Event')

plt.xlabel('Tanggal Event')
plt.ylabel('Jumlah')

plt.show()
```



Dari grafik dapat dilihat bahwa pengguna berangsur naik dan menurun setelah tanggal 21 yang mana menjadi tanggal puncak.

A/B Testing

Nilai dari hipotesis nol (H_0) dan hipotesis alternatif (H_1) harus ditentukan terlebih dahulu, yaitu sebagai berikut

- H_0 : Secara statistik jumlah sampel A dan B adalah sama.
- H_1 : Secara statistik jumlah sampel A dan B adalah tidak sama.

Nilai alpha yang digunakan adalah 5%, dan metode uji yang digunakan adalah z-test.

```
In [69]: user_count = df.groupby('group').agg({'user_id': 'nunique'})
         user_count
```

```
Out[69]:
```

	user_id
group	
A	2604
B	877

```
In [70]: event_user_count = df.pivot_table(
         index='event_name',
         columns='group',
         values='user_id',
         aggfunc='nunique').sort_values(by='A', ascending=False)

         event_user_count
```

```
Out[70]:
```

	group	A	B
event_name			
login		2604	876
product_page		1685	493
purchase		833	249
product_cart		782	244

Selanjutnya akan dibuat fungsi untuk menguji hipotesis.

```
In [71]: def check_hypothesis(successes1, successes2, trials1, trials2, alpha):
         p1 = successes1/trials1

         p2 = successes2/trials2

         p_combined = (successes1 + successes2) / (trials1 + trials2)

         difference = p1 - p2

         z_value = difference / math.sqrt(p_combined * (1 - p_combined) * (1/trials1 + 1/trials2))

         distr = st.norm(0, 1)

         p_value = (1 - distr.cdf(abs(z_value))) * 2

         print('p-value: ', p_value)

         if (p_value < alpha):
             print("Kita menolak hipotesis nol")
```

```

else:
    print("Kita tidak bisa mengabaikan hipotesis nol")

```

```

In [72]: def check_event_hypothesis(df_event, df_all, event, exp1, exp2, col, alpha):

    frac1 = (df_event.loc[event, exp1] / df_all.loc[exp1])*100
    frac2 = (df_event.loc[event, exp2] / df_all.loc[exp2])*100

    print(f'Persentase peristiwa: {round(frac1.loc[col], 2)}%, peristiwa {event}, {exp1}')
    print(f'Persentase peristiwa: {round(frac2.loc[col], 2)}%, peristiwa {event}, {exp2}')
    print()

    check_hypothesis(df_event.loc[event, exp1],
                     df_event.loc[event, exp2],
                     df_all.loc[exp1],
                     df_all.loc[exp2],
                     alpha)

    print('-----')

```

```

In [73]: for event in event_user_count.index:
    check_event_hypothesis(event_user_count, user_count, event, exp1='A', exp2='B')
    print()

```

```

Persentase peristiwa: 100.0%, peristiwa login, group A
Persentase peristiwa: 99.89%, peristiwa login, group B

```

```

p-value: [0.08481837]
Kita menolak hipotesis nol

```

```

-----

Persentase peristiwa: 64.71%, peristiwa product_page, group A
Persentase peristiwa: 56.21%, peristiwa product_page, group B

```

```

p-value: [6.94273936e-06]
Kita menolak hipotesis nol

```

```

-----

Persentase peristiwa: 31.99%, peristiwa purchase, group A
Persentase peristiwa: 28.39%, peristiwa purchase, group B

```

```

p-value: [0.04652483]
Kita menolak hipotesis nol

```

```

-----

Persentase peristiwa: 30.03%, peristiwa product_cart, group A
Persentase peristiwa: 27.82%, peristiwa product_cart, group B

```

```

p-value: [0.21469192]
Kita menolak hipotesis nol

```

Berdasarkan hasil A/B Testing yang diberikan, dapat dikatakan bahwa terdapat perbedaan yang signifikan proporsi transaksi grup A dan B untuk transaksi login, product page, dan purchase. Kita menolak hipotesis nol bahwa tidak ada perbedaan antara kelompok A dan B untuk ketiga peristiwa ini. Namun, tidak ada perbedaan signifikan yang ditemukan antara grup A dan B untuk event product cart. Kita tidak dapat menolak hipotesis nol bahwa tidak ada perbedaan antara kelompok A dan B dalam kasus ini.

Kesimpulan

Dari hasil pengolahan data diatas didapatkan beberapa kesimpulan, yaitu:

- Terdapat perbedaan persentase transaksi yang signifikan antara kelompok A dan B untuk transaksi login, product page dan purchase.
- Hal ini menunjukkan bahwa perubahan yang dilakukan pada sistem rekomendasi mempengaruhi perilaku pengguna pada tahap awal dan akhir corong.